

Yuzhen Song

Southern University of Science and Technology

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🐱 Personal Page • 🌐 GitHub Repository

Research Interests: Humanoid Robotics; Robot Learning; Embodied Intelligence.

EDUCATION

Southern University of Science and Technology

B.Eng. in Robotics Engineering

Shenzhen, China

Aug 2023 – Jun 2027

- **Overall GPA: 3.87/4.0** (Weighted Avg: **93.03/100**) **Major GPA: 3.94/4.0**
- **Core Courses:** Robot Operating System (100), Mechanics of Materials M (100), Mechatronic Systems (98), Robot Modeling and Control (97), Mechanical Design (97), Finite Element Theory (97), Intro to C Programming (97), Signals and Systems (95), Sensing Technology (95), Linear Algebra (94), Machine Learning for Eng. (93), Robotic Actuation System (93), Computer Vision (90), Fundamentals of Control Eng. (90)
- **Leadership:** President of Robotics Club & Student Organizations Council (SOC)

University of California, Berkeley

Visiting Student

Berkeley, CA, USA

Jul 2026 – Dec 2026

PUBLICATIONS

- J. Cao*, Y. Chen*, **Y. Song**, M. Tomizuka, C. Li, T. Tian, “**TEXEDO: Test Time Scaling for Controller-Aware Language-Conditioned Humanoid Motion Generation**,” *Conference on Robot Learning (CoRL)*, 2026. **Accepted**. Also presented at the WCBM and RoboData workshops at RSS 2026.

RESEARCH EXPERIENCES

SignHOI – Human-to-Robot HOI Retargeting for Humanoid Manipulation

University of California, Berkeley

Berkeley, CA, USA

Aug 2026 – Present

Visiting Undergraduate Researcher; Advisor: Prof. Masayoshi Tomizuka

- Developing a pipeline to retarget generated human-object interaction motions to humanoid robots, with a focus on fine-grained hand and manipulation behaviors.
- Investigating VLM-based trajectory verification and correction together with reinforcement learning to improve physical feasibility and real-robot execution success.

TEXEDO – Test-Time Scaling for Controller-Aware Humanoid Motion Generation

University of California, Berkeley

Apr 2026 – Sept 2026

Visiting Undergraduate Researcher; Advisor: Prof. Masayoshi Tomizuka

- Integrated Kimodo motions into MuJoCo via mlab, aligning quaternion conventions, sampling rates, and evaluation logs.
- Compared TEXEDO with Kimodo using MPJE, velocity errors, and acceleration errors; resolved memory failures through streaming computation and logging.
- Built a Gemini video evaluator for semantic alignment and physical plausibility, comparing automated scores with manual ratings.
- Co-authored TEXEDO, accepted to **CoRL 2026** and presented at the RSS 2026 WCBM and RoboData workshops.

A Model for Predicting Metabolically Optimal Anteroposterior Center-of-Mass Assistance During Uphill Walking

Southern University of Science and Technology

Shenzhen, China

Sep 2024 – Present

Undergraduate Researcher; Advisor: Prof. Chenglong Fu

- Reproduced the team's equations for single and double support and implemented code to predict and plot optimal assistance across slopes.
- Designed a cable assistance platform with a motorized spool, a rail and pulley mechanism for angle adjustment, and a waist belt attachment.
- Participating in ongoing treadmill experiments to evaluate metabolic responses to predicted assistance.

Simulation of an Underactuated Metamorphic Palm for a Three-Fingered Hand

Southern University of Science and Technology

Jun 2025 – Oct 2025

Undergraduate Researcher; Advisors: Prof. Jiansheng Dai, Prof. Zhao Tang

- Built Isaac Sim and Webots environments for an underactuated palm with five links, adding closure constraints to URDF models for preliminary RL experiments.
- Simplified meshes while preserving geometry at contact regions, resolving GPU memory failures and enabling simulation on an 8 GB GPU.
- Investigated instability during palm reconfiguration by analyzing joint constraints and tuning damping, stiffness, and friction, identifying limitations in simulation fidelity.

SELECTED PROJECTS

BEAT – Barometric Elevation and Attitude Tracking for Posture-Independent Wrist Blood Pressure Measurement

Shenzhen, China

Southern University of Science and Technology

Mar 2026 – Jun 2026

Project Leader; Advisor: Prof. Chengzhi Hu

- Led BEAT development, using barometric sensing to correct for wrist height relative to the heart.
- Programmed two ESP32 boards for ESP-NOW communication and built a desktop UI for live sensor monitoring.
- Designed and analyzed pilot tests across three arm positions ($n = 2$); compensation reduced systolic MAE by **56.2%**.

SplatSLAM – Indoor Reconstruction with SLAM and 3D Gaussian Splatting

Southern University of Science and Technology

Mar 2026 – Jun 2026

Core Team Member; Advisor: Prof. Feng Zheng

- Integrated MAST3R-SLAM with Nerfstudio’s 3DGS pipeline to generate point clouds, rendered images, and walkthrough videos from monocular video.
- Analyzed how training iterations affected PSNR, L1 loss, and visual quality, identifying signs of overfitting and spiky rendering artifacts.

ROS 2 Mobile Manipulator for Navigation and Grasping

Southern University of Science and Technology

Sep 2025 – Dec 2025

Project Leader; Course Advisor: Prof. Wende Ke

- Integrated slam_toolbox, Nav2, and EKF fusion of wheel odometry and IMU data for mapping, localization, and navigation in Gazebo.
- Integrated OpenCV visual alignment and arm control for object retrieval, with voice and gesture interfaces using Vosk and MediaPipe.

Image Processing and Motion Planning for Robotic Drawing

Southern University of Science and Technology

Mar 2025 – Jun 2025

Project Leader; Course Advisor: Prof. Chenglong Fu

- Developed OpenCV routines for contour extraction and skeletonization to generate drawing paths for a six-axis robotic arm.
- Used inverse kinematics in MATLAB to convert planar drawing paths into joint trajectories and verified the resulting motion in simulation.
- Implemented tool orientation adjustment along surface normals for simulated drawing on cylindrical surfaces.

Autonomous Rescue Robot (National Competition)

Southern University of Science and Technology

Dec 2024 – Aug 2025

Project Leader; Advisors: Prof. Dong Lu, Yuanjian Huang

- Led the development of a mobile robot integrating YOLOv11 object detection and STM32 motion control for autonomous target tracking and retrieval.
- Won **National Second Prize** and **Provincial First Prize** in the China Undergraduate Engineering Practice and Innovation Ability Competition.

SKILLS

- **Programming:** C/C++, Python, MATLAB
- **Robotics & Simulation:** ROS 2, MuJoCo, Isaac Sim, Isaac Lab, Gazebo, Webots
- **Computer Vision & Visualization:** OpenCV, RViz2
- **Embedded Systems:** STM32, ESP32
- **CAD/CAM & CAE:** SolidWorks, AutoCAD, Edgcam, Abaqus

HONORS & AWARDS

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| ◦ National Second Prize, China Undergraduate Eng. Practice and Innovation Ability Competition | 2025 |
| ◦ Provincial First Prize, Guangdong Undergraduate Eng. Practice and Innovation Ability Competition | 2025 |
| ◦ First-Class Scholarship for Outstanding Students (Top 3%) | 2025 |
| ◦ Second-Class Scholarship for Outstanding Students (Top 8%) | 2024, 2026 |
| ◦ Outstanding Student Award | 2024, 2025, 2026 |
| ◦ Provincial Second Prize, Guangdong Undergraduate Mechanical Innovational Competition | 2025 |